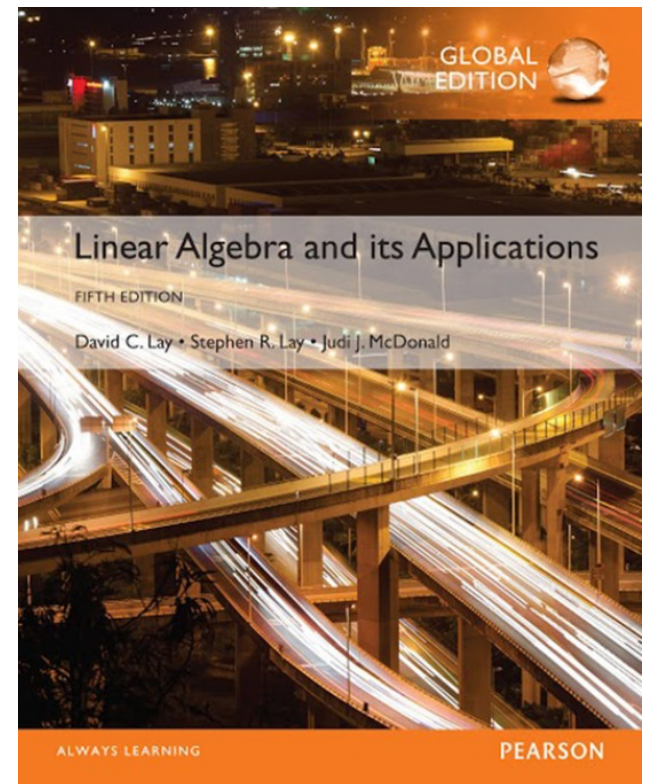


3 Determinants

3.1

INTRODUCTION TO DETERMINANTS



INTRODUCTION TO DETERMINANTS

- **Definition:** For $n \geq 2$, the **determinant** of an $n \times n$ matrix $A = [a_{ij}]$ is the sum of n terms of the form $\pm a_{1j} \det A_{1j}$, with plus and minus signs alternating, where the entries $a_{11}, a_{12}, \dots, a_{1n}$ are from the first row of A . In symbols,

$$\begin{aligned}\det A &= a_{11} \det A_{11} - a_{12} \det A_{12} + \cdots + (-1)^{1+n} a_{1n} \det A_{1n} \\ &= \sum_{j=1}^n (-1)^{1+j} a_{1j} \det A_{1j}\end{aligned}$$

INTRODUCTION TO DETERMINANTS

- **Example 1** Compute the determinant of

$$A = \begin{bmatrix} 1 & 5 & 0 \\ 2 & 4 & -1 \\ 0 & -2 & 0 \end{bmatrix}$$

- **Solution** Compute $\det A = a_{11}\det A_{11} - a_{12}\det A_{12} + a_{13}\det A_{13}$:

$$\det A = 1 \cdot \det \begin{bmatrix} 4 & -1 \\ -2 & 0 \end{bmatrix} - 5 \cdot \det \begin{bmatrix} 2 & -1 \\ 0 & 0 \end{bmatrix} + 0 \cdot \det \begin{bmatrix} 2 & 4 \\ 0 & -2 \end{bmatrix}$$

$$= 1(0 - 2) - 5(0 - 0) + 0(-4 - 0) = -2$$

INTRODUCTION TO DETERMINANTS

- Another common notation for the determinant of a matrix uses a pair of vertical lines in place of brackets.
- Thus the calculation in Example 1 can be written as

$$\det A = 1 \cdot \begin{vmatrix} 4 & -1 \\ -2 & 0 \end{vmatrix} - 5 \begin{vmatrix} 2 & -1 \\ 0 & 0 \end{vmatrix} + 0 \begin{vmatrix} 2 & 4 \\ 0 & -2 \end{vmatrix} = \dots = -2$$

- To state the next theorem, it is convenient to write the definition of $\det A$ in a slightly different form. Given $A = [a_{ij}]$, the **(i, j)-cofactor** of A is the number C_{ij} given by

$$C_{ij} = (-1)^{i+j} \det A_{ij}. \quad (4)$$

INTRODUCTION TO DETERMINANTS

- Then

$$\det A = a_{11}C_{11} + a_{12}C_{12} + \cdots + a_{1n}C_{1n}$$

- This formula is called a **cofactor expansion across the first row** of A .

- **Theorem 1:** The determinant of an $n \times n$ matrix A can be computed by a cofactor across any row or down any column. The expansion across the i th row using the cofactors in (4) is

$$\det A = a_{i1}C_{i1} + a_{i2}C_{i2} + \cdots + a_{in}C_{in}$$

- The cofactor expansion down the j th column is

$$\det A = a_{1j}C_{1j} + a_{2j}C_{2j} + \cdots + a_{nj}C_{nj}$$

INTRODUCTION TO DETERMINANTS

- **Example 2** Use a cofactor expansion across the third row to compute $\det A$, where

$$A = \begin{bmatrix} 1 & 5 & 0 \\ 2 & 4 & -1 \\ 0 & -2 & 0 \end{bmatrix}$$

- **Solution** Compute

$$\begin{aligned} \det A &= a_{31}C_{31} + a_{32}C_{32} + a_{33}C_{33} \\ &= (-1)^{3+1}a_{31}\det A_{31} + (-1)^{3+2}a_{32}\det A_{32} + (-1)^{3+3}a_{33}\det A_{33} \\ &= \begin{vmatrix} 5 & 0 \\ 4 & -1 \end{vmatrix} - (-2)\begin{vmatrix} 1 & 0 \\ 2 & -1 \end{vmatrix} + 0\begin{vmatrix} 1 & 5 \\ 2 & 4 \end{vmatrix} \\ &= 0 + 2(-1) + 0 = -2 \end{aligned}$$

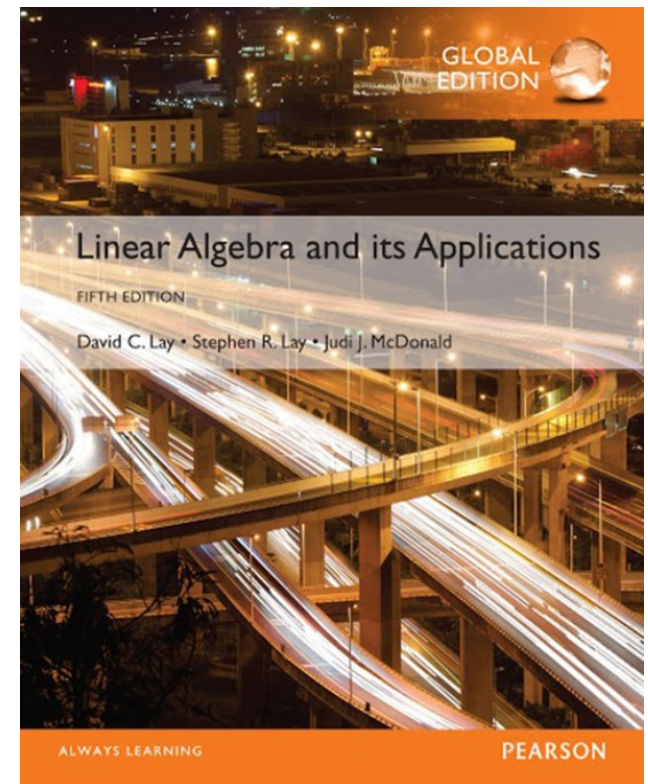
INTRODUCTION TO DETERMINANTS

- **Theorem 2:** If A is a triangular matrix, then $\det A$ is the product of the entries on the main diagonal of A .

3 Determinants

3.2

PROPERTIES OF DETERMINANTS



PROPERTIES OF DETERMINANTS

- **Theorem 3:** Let A be a square matrix
 - a) If a multiple of one row of A is added to another row to produce a matrix B , then $\det B = \det A$.
 - b) If two rows of A are interchanged to produce B , then $\det B = -\det A$.
 - c) If one row of A is multiplied by k to produce B , then $\det B = k \cdot \det A$

PROPERTIES OF DETERMINANTS

- **Example 1** Compute $\det A$, where $A = \begin{bmatrix} 1 & -4 & 2 \\ -2 & 8 & -9 \\ -1 & 7 & 0 \end{bmatrix}$
- **Solution** The strategy is to reduce A to echelon form and then to use the fact that the determinant of a triangular matrix is the product of the diagonal entries. The first two row replacements in column 1 do not change the determinant:

$$\det A = \begin{vmatrix} 1 & -4 & 2 \\ -2 & 8 & -9 \\ -1 & 7 & 0 \end{vmatrix} = \begin{vmatrix} 1 & -4 & 2 \\ 0 & 0 & -5 \\ -1 & 7 & 0 \end{vmatrix} = \begin{vmatrix} 1 & -4 & 2 \\ 0 & 0 & -5 \\ 0 & 3 & 2 \end{vmatrix}$$

PROPERTIES OF DETERMINANTS

- An interchange of rows 2 and 3 reverses the sign of the determinant, so

$$\det A = - \begin{vmatrix} 1 & -4 & 2 \\ 0 & 3 & 2 \\ 0 & 0 & -5 \end{vmatrix} = -(1)(3)(-5) = 15$$

PROPERTIES OF DETERMINANTS

- **Theorem 4:** A square matrix A is invertible if and only if $\det A \neq 0$.

- **Example 3** Compute $\det A$, where $A = \begin{bmatrix} 3 & -1 & 2 & -5 \\ 0 & 5 & -3 & -6 \\ -6 & 7 & -7 & 4 \\ -5 & -8 & 0 & 9 \end{bmatrix}$

- **Solution** Add 2 times row 1 to row 3 to obtain

$$\det A = \det \begin{bmatrix} 3 & -1 & 2 & -5 \\ 0 & 5 & -3 & -6 \\ 0 & 5 & -3 & -6 \\ -5 & -8 & 0 & 9 \end{bmatrix} = 0$$

because the second and third rows of the second matrix are equal.

COLUMN OPERATIONS

- **Theorem 5:** If A is a $n \times n$ matrix, then $\det A^T = \det A$.
- **Proof:** The theorem is obvious for $n = 1$. Suppose the theorem is true for $k \times k$ determinants and let $n = k + 1$.
- Then the cofactor of a_{1j} in A equals the cofactor of a_{j1} in A^T , because the cofactors involve $k \times k$ determinants.
- Hence the cofactor expansion of $\det A$ along the first row equals the cofactor expansion of $\det A^T$ down the first column. That is, A and A^T have equal determinants.
- Thus the theorem is true for $n = 1$, and the truth of the theorem for one value of n implies its truth for the next value of n . By the principle of induction, the theorem is true for all $n \geq 1$.

DETERMINANTS AND MATRIX PRODUCTS

- **Theorem 6:** If A and B are $n \times n$ matrices, then $\det AB = (\det A)(\det B)$.

- **Example 5** Verify Theorem 6 for $A = \begin{bmatrix} 6 & 1 \\ 3 & 2 \end{bmatrix}$ and $B = \begin{bmatrix} 4 & 3 \\ 1 & 2 \end{bmatrix}$.

- Solution

$$AB = \begin{bmatrix} 6 & 1 \\ 3 & 2 \end{bmatrix} \begin{bmatrix} 4 & 3 \\ 1 & 2 \end{bmatrix} = \begin{bmatrix} 25 & 20 \\ 14 & 13 \end{bmatrix}$$

- and

$$\det AB = 25 \cdot 13 - 20 \cdot 14 = 325 - 280 = 45$$

Since $\det A = 9$ and $\det B = 5$,

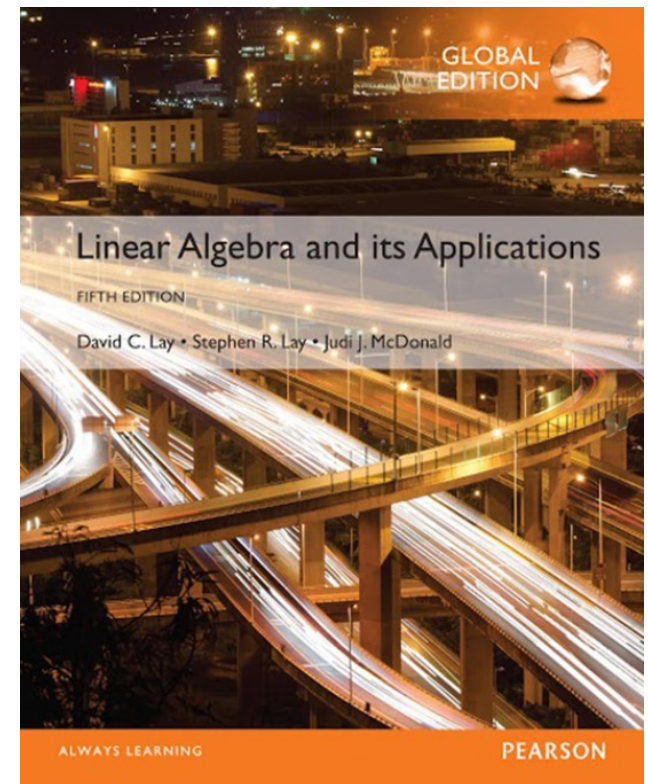
$$(\det A)(\det B) = 9 \cdot 5 = 45 = \det AB$$

3

Determinants

3.3

CRAMER'S RULE, VOLUME, AND LINEAR TRANSFORMATIONS



CRAMER'S RULE

- **Theorem 7:** Let A be an invertible $n \times n$ matrix. For any b in $\mathbb{R}^n$, the unique solution x of $Ax=b$ has entries given by

$$x_i = \frac{\det A_i(b)}{\det A}, \quad i = 1, 2, \dots, n \quad (1)$$

- **Proof** Denote the columns of A by $a_1, \dots, a_n$ and the columns of the $n \times n$ identity matrix I by $e_1, \dots, e_n$. If $Ax = b$, the definition of matrix multiplication shows that

$$\begin{aligned} A \cdot Ii(x) &= A[e_1 \quad \dots \quad x \quad \dots \quad e_n] = A[e_1 \quad \dots \quad Ax \quad \dots \quad Ae_n] \\ &= [a_1 \quad \dots \quad b \quad \dots \quad a_n] = A_i(b) \end{aligned}$$

CRAMER'S RULE

- By the multiplicative property of determinants,

$$(\det A)(\det I_i(x)) = \det A_i(b)$$

- The second determinant on the left is simply x_i . Hence $(\det A) \cdot x_i = \det A_i(b)$. This proves (1) because A is invertible and $\det A \neq 0$.

- **Example 1** Use Cramer's rule to solve the system

$$\begin{aligned} 3x_1 - 2x_2 &= 6 \\ -5x_1 + 4x_2 &= 8 \end{aligned}$$

CRAMER'S RULE

- **Solution** View the system as $Ax = b$. Using the notation introduced above,

$$A = \begin{bmatrix} 3 & -2 \\ -5 & 4 \end{bmatrix}, \quad A_1(b) = \begin{bmatrix} 6 & -2 \\ 8 & 4 \end{bmatrix}, \quad A_2(b) = \begin{bmatrix} 3 & 6 \\ -5 & 8 \end{bmatrix}$$

- Since $\det A = 2$, the system has a unique solution. By Cramer's rule,

$$x_1 = \frac{\det A_1(b)}{\det A} = \frac{24 + 16}{2} = 20$$

$$x_2 = \frac{\det A_2(b)}{\det A} = \frac{24 + 30}{2} = 27$$

A FORMULA FOR A^{-1}

- **Theorem 8:** Let A be an invertible $n \times n$ matrix. Then

$$A^{-1} = \frac{1}{\det A} \text{adj} A$$

- **Example 3** Find the inverse of the matrix $A = \begin{bmatrix} 2 & 1 & 3 \\ 1 & -1 & 1 \\ 1 & 4 & -2 \end{bmatrix}$.

- **Solution** The nine cofactors are

$$C_{11} = + \begin{vmatrix} -1 & 1 \\ 4 & -2 \end{vmatrix} = -2, \quad C_{12} = - \begin{vmatrix} 1 & 1 \\ 1 & -2 \end{vmatrix} = 3, \quad C_{13} = + \begin{vmatrix} 1 & -1 \\ 1 & 4 \end{vmatrix} = 5$$

$$C_{21} = - \begin{vmatrix} 1 & 3 \\ 4 & -2 \end{vmatrix} = 14, \quad C_{22} = + \begin{vmatrix} 2 & 3 \\ 1 & -2 \end{vmatrix} = -7, \quad C_{23} = - \begin{vmatrix} 2 & 1 \\ 1 & 4 \end{vmatrix} = -7$$

$$C_{31} = + \begin{vmatrix} 1 & 1 \\ -1 & 1 \end{vmatrix} = 4, \quad C_{32} = - \begin{vmatrix} 2 & 3 \\ 1 & 1 \end{vmatrix} = 1, \quad C_{33} = + \begin{vmatrix} 2 & 1 \\ 1 & -1 \end{vmatrix} = -3$$

A FORMULA FOR A^{-1}

- The adjugate matrix is the *transpose* of the matrix of cofactors. Thus

$$\text{adj}A = \begin{bmatrix} -2 & 14 & 4 \\ 3 & -7 & 1 \\ 5 & -7 & -3 \end{bmatrix}$$

- We could compute $\det A$ directly, but the following computation provides a check on the calculations above and produces $\det A$:

$$(\text{adj}A) \cdot A = \begin{bmatrix} -2 & 14 & 4 \\ 3 & -7 & 1 \\ 5 & -7 & -3 \end{bmatrix} \begin{bmatrix} 2 & 1 & 3 \\ 1 & -1 & 1 \\ 1 & 4 & -2 \end{bmatrix} = 14I$$

A FORMULA FOR A^{-1}

- Since $(\text{adj } A)A = 14I$, Theorem 8 shows that $\det A = 14$ and

$$A^{-1} = \frac{1}{14} \begin{bmatrix} -2 & 14 & 4 \\ 3 & -7 & 1 \\ 5 & -7 & -3 \end{bmatrix} = \begin{bmatrix} -1/7 & 1 & 2/7 \\ 3/14 & -1/2 & 1/14 \\ 5/14 & -1/2 & -3/14 \end{bmatrix}$$

DETERMINANTS AS AREA OR VOLUME

- **Theorem 9:** If A is a 2×2 matrix, the area of the parallelogram determined by the columns of A is $|\det A|$. If A is a 3×3 matrix, the volume of the parallelepiped determined by the columns of A is $|\det A|$.
- **Proof** The theorem is obviously true for any 2×2 diagonal matrix:
$$\left| \det \begin{bmatrix} a & 0 \\ 0 & d \end{bmatrix} \right| = |ad| = \left\{ \begin{array}{l} \text{area of} \\ \text{rectangle} \end{array} \right\}$$
- See Fig. 1 on the next slide.

DETERMINANTS AS AREA OR VOLUME

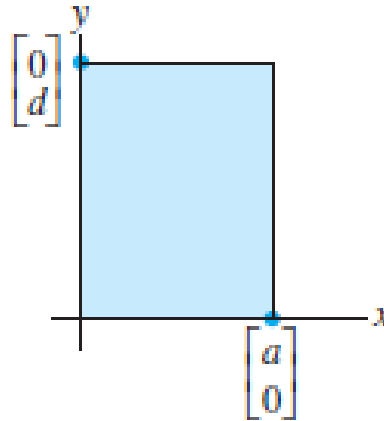


FIGURE 1

$$\text{Area} = |ad|.$$

- It will suffice to show that any 2×2 matrix $A = [a_1 a_2]$ can be transformed into a diagonal matrix in a way that changes neither the area of the associated parallelogram nor $|\det A|$.

DETERMINANTS AS AREA OR VOLUME

- It suffices to prove the following simple geometric observation that applies to vectors in $\mathbb{R}^2$ or $\mathbb{R}^3$:
- Let a_1 and a_2 be nonzero vectors. Then for any scalar c , the area of the parallelogram determined by a_1 and a_2 equals the area of the parallelogram determined by a_1 and $a_2 + ca_1$.
- To prove this statement, we may assume that a_2 is not a multiple of a_1 , for otherwise the two parallelograms would be degenerate and have zero area.
- If L is the line through 0 and a_1 , then $a_2 + L$ is the line through a_2 parallel to L , and $a_2 + ca_1$ is on this line. See Fig. 2 on the next slide.

DETERMINANTS AS AREA OR VOLUME

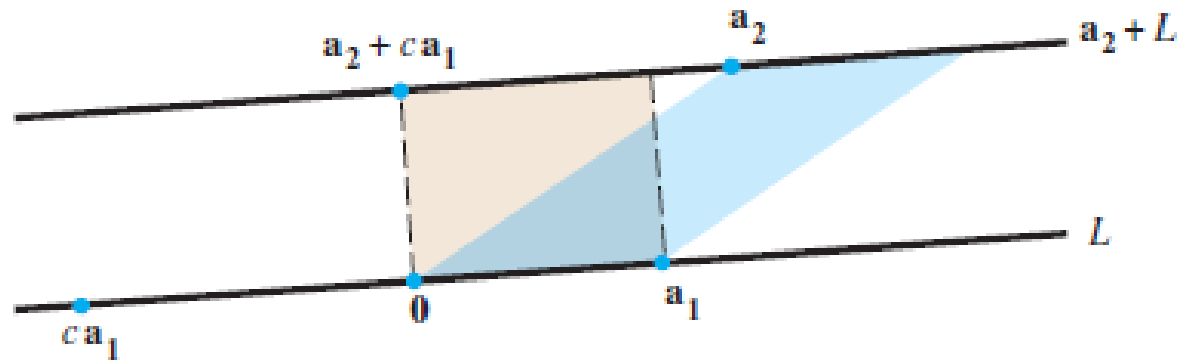


FIGURE 2 Two parallelograms of equal area.

- The points a_2 and $a_2 + ca_1$ have the same perpendicular distance to L . Hence the two parallelograms in Fig. 2 have the same area, since they share the base from 0 to a_1 .

DETERMINANTS AS AREA OR VOLUME

- The proof for $\mathbb{R}^3$ is similar. The theorem is obviously true for a 3×3 diagonal matrix. See Fig. 3 below:

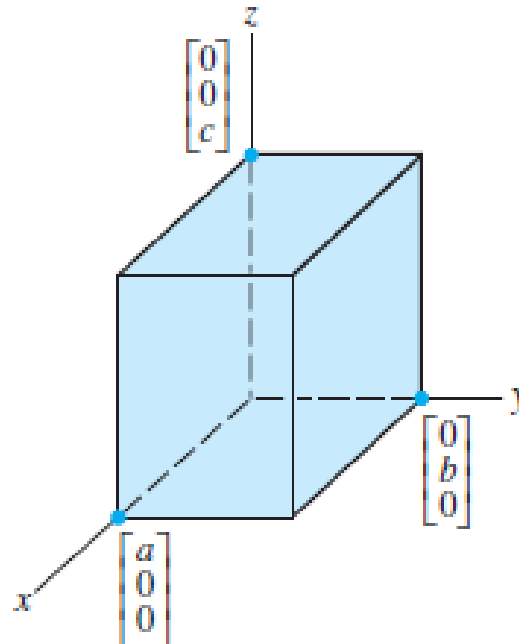


FIGURE 3

$$\text{Volume} = |abc|.$$

DETERMINANTS AS AREA OR VOLUME

- And any 3×3 matrix A can be transformed into a diagonal matrix using column operations that do not change $|\det A|$.
- A parallelepiped is shown in Fig. 4 below as a shaded box with two sloping sides.

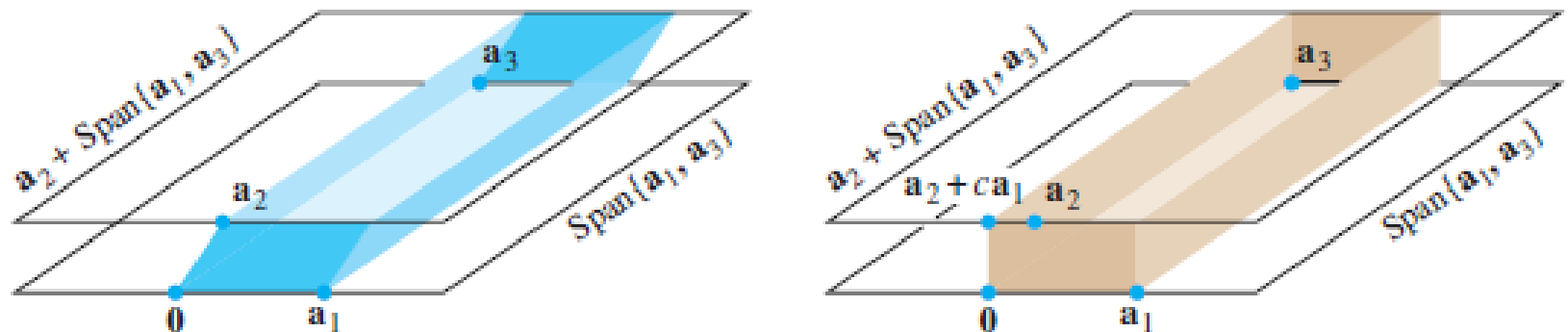


FIGURE 4 Two parallelepipeds of equal volume.

DETERMINANTS AS AREA OR VOLUME

- Its volume is the area of the base in the plane $\text{Span}\{a_1, a_3\}$ times the altitude of a_2 above $\text{Span}\{a_1, a_3\}$. Any vector $a_2 + ca_1$ lies in the plane $\text{Span}\{a_1, a_3\}$, which is parallel to $\text{Span}\{a_1, a_3\}$.
- Hence the volume of the parallelepiped is unchanged when $[a_1 \ a_2 \ a_3]$ is changed to $[a_1 \ a_2 + ca_1 \ a_3]$.
- Thus a column replacement operation does not affect the volume of the parallelepiped. Since the column interchanges have no effect on the volume, the proof is complete.

DETERMINANTS AS AREA OR VOLUME

- **Example 4** Calculate the area of the parallelogram determined by the points $(-2, -2)$, $(0, 3)$, $(4, -1)$, and $(6, 4)$. See Fig. 5(a) below:

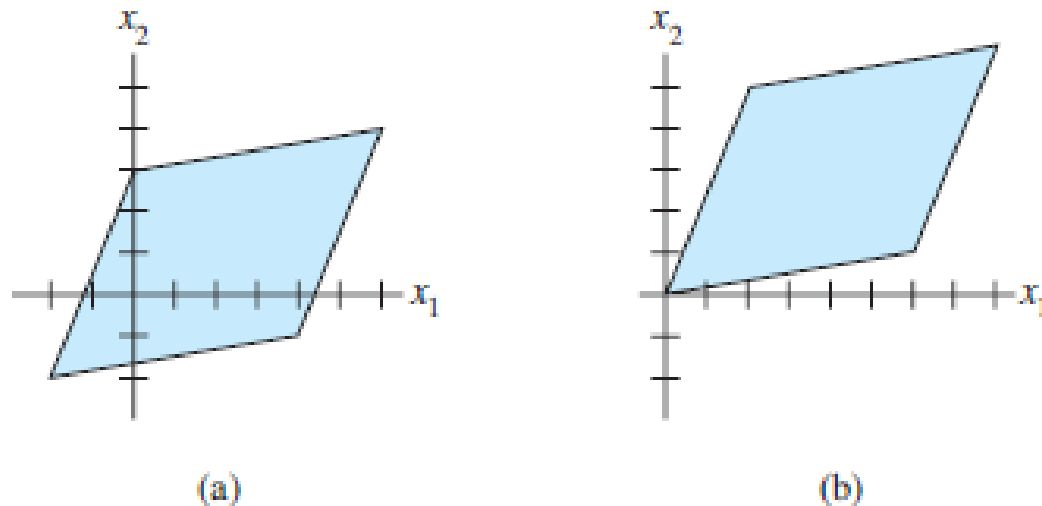


FIGURE 5 Translating a parallelogram does not change its area.

DETERMINANTS AS AREA OR VOLUME

- **Solution** First translate the parallelogram to one having the origin as a vertex. For example, subtract the vertex $(-2, -2)$ from each of the four vertices.
- The new parallelogram has the same area, and its vertices are $(0, 0)$, $(2, 5)$, $(6, 1)$, and $(8, 6)$. See Fig. 5(b) on the previous slide.
- This parallelogram is determined by the columns of
$$A = \begin{bmatrix} 2 & 6 \\ 5 & 1 \end{bmatrix}$$
- Since $|\det A| = |-28|$, the area of the parallelogram is 28.

LINEAR TRANSFORMATIONS

- **Theorem 10:** Let $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be the linear transformation determined by a 2×2 matrix A . If S is a parallelogram in $\mathbb{R}^2$, then

$$\{\text{area of } T(S)\} = |\det A| \cdot \{\text{area of } S\} \quad (5)$$

- If T is determined by a 3×3 matrix A , and if S is a parallelepiped in $\mathbb{R}^3$, then

$$\{\text{volume of } T(S)\} = |\det A| \cdot \{\text{volume of } S\} \quad (6)$$

- **Proof** Consider the 2×2 case, with $A = [a_1 \ a_2]$. A parallelogram at the origin in $\mathbb{R}^2$ determined by vectors b_1 and b_2 has the form

$$S = \{s_1 b_1 + s_2 b_2: 0 \leq s_1 \leq 1, 0 \leq s_2 \leq 1\}$$

LINEAR TRANSFORMATIONS

- The image of S under T consists of points of the form
$$\begin{aligned}T(s_1b_1 + s_2b_2) &= s_1T(b_1) + s_2T(b_2) \\ &= s_1Ab_1 + s_2Ab_2\end{aligned}$$
- where $0 \leq s_1 \leq 1$, $0 \leq s_2 \leq 1$. It follows that $T(S)$ is the parallelogram determined by the columns of the matrix $[Ab_1 \ Ab_2]$. This matrix can be written as AB , where $B = [b_1 \ b_2]$.
- By Theorem 9 and the product theorem for determinants,
$$\begin{aligned}\{area\ of\ T(S)\} &= |detAB| = |detA| \cdot |detB| \\ &= |detA| \cdot \{area\ of\ S\}\end{aligned}\tag{7}$$

LINEAR TRANSFORMATIONS

- An arbitrary parallelogram has the form $\mathbf{p} + S$, where $\mathbf{p}$ is a vector and S is a parallelogram at the origin.

- It is easy to see that T transforms $\mathbf{p} + S$ into $T(\mathbf{p}) + T(S)$. Since translation does not affect the area of a set,

$$\begin{aligned}\{\text{area of } T(\mathbf{p} + S)\} &= \{\text{area of } T(\mathbf{p}) + T(S)\} \\ &= \{\text{area of } T(S)\}\end{aligned}$$

Translation

$$= |\det A| \cdot \{\text{area of } S\}$$

By equation (7)

$$= |\det A| \cdot \{\text{area of } \mathbf{p} + S\}$$

Translation

- This shows that (5) holds for all parallelograms in $\mathbb{R}^2$. The proof of (6) for the 3×3 case is analogous.

LINEAR TRANSFORMATIONS

- **Example 5** Let a and b be positive numbers. Find the area of the region E bounded by the ellipse whose equation is

$$\frac{x_1^2}{a^2} + \frac{x_2^2}{b^2} = 1$$

- **Solution** We claim that E is the image of the unit disk D under the linear transformation T determined by the matrix $A = \begin{bmatrix} a & 0 \\ 0 & b \end{bmatrix}$, because if $\mathbf{u} = \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}$, $\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$, and $\mathbf{x} = A\mathbf{u}$, then

$$u_1 = \frac{x_1}{a} \text{ and } u_2 = \frac{x_2}{b}$$

LINEAR TRANSFORMATIONS

- It follows that $\mathbf{u}$ is in the unit disk, with $u_1^2 + u_2^2 \leq 1$, if and only if $\mathbf{x}$ is in E , with $(x_1/a)^2 + (x_2/b)^2 \leq 1$. By generalization of Theorem 10,

$$\begin{aligned}\{\text{area of ellipse}\} &= \{\text{area of } T(D)\} \\ &= |\det A| \cdot \{\text{area of } D\} \\ &= ab \cdot \pi(1)^2 = \pi ab\end{aligned}$$